Ink Density Saturation in Duan Inkstones

Anonymous submission

Abstract

A minimal document in the AAAI 2027 submission format. The content mirrors `latex-rendering-check.tex`: ink density saturates as a function of grind time, and the stone — not the stick — sets the ceiling. We formalize the saturation dynamics as a first-order process, report fitted parameters for three historically significant stones, and argue that the grinding surface, rather than the ink stick, determines both the rate of saturation and the achievable ceiling. The document doubles as a rendering check: it exercises the running head, numbered equations, booktabs tables, and natbib citations under the AAAI class.

Introduction

The preparation of liquid ink from a solid ink stick is one of the oldest continuously practiced material processes in East Asian scholarship. A stick of compressed soot and animal glue is ground against a wetted stone until the resulting suspension reaches a working density. Practitioners have long claimed that the choice of stone matters more than the choice of stick, a claim recorded as early as the classical treatises surveyed by Mizuta (1968). Yet the claim was largely anecdotal until modern densitometry made it testable.

This note formalizes the folk claim as a parameter-identification problem. We model grinding as a first-order saturating process, fit its two parameters to published measurements, and compare stones from the two most celebrated quarry traditions: Duan and She. Our contribution is deliberately modest — a clean restatement of known results in a form suitable for quantitative comparison — but the framing clarifies which physical properties of a stone are load-bearing and which are cosmetic.

Related Work

The empirical study of inkstone performance is small but well documented. Tanaka and Sato (2019) established the saturating-exponential form used here by grinding standardized pine-soot sticks on instrumented stones and sampling optical density at fixed intervals. Their measurements cover the Laokeng and Majiakeng quarries of the Duan tradition alongside She stones from Longwei mountain. Chen and Nakamura (2021) approach the same materials from the provenance side, surveying spectroscopic and petrographic methods for attributing a stone to a quarry; their taxonomy of surface microstructure informs the discussion in the final section. The broader cultural context of the scholar's four treasures — stone, stick, brush, and paper — is treated at length by Mizuta (1968).

Model

Let $D(t)$ denote ink density after t minutes of grinding. Density follows a saturating exponential,

$$D(t) = D_{\max}\left(1 - e^{-t/\tau}\right), \tag{1}$$

where τ is the stone's time constant and $D_{\max}$ its ceiling. The relationship was established empirically by Tanaka and Sato (2019).

Two properties of Equation 1 deserve emphasis. First, the model has no parameter describing the ink stick: within the range of sticks tested, stick composition shifts neither τ nor $D_{\max}$ beyond measurement error. Second, the two parameters are identifiable from a short grinding session, since

$$\left.\frac{dD}{dt}\right|_{t=0} = \frac{D_{\max}}{\tau} \tag{2}$$

gives the initial slope and any two well-separated samples then pin both values. In practice a calligrapher senses Equation 2 directly: a fast stone announces itself within the first minute.

The physical interpretation is straightforward. The stone's surface carries microscopic abrasive structure — the "forge" patterns of the quarry — which shears particles off the stick and disperses them into the water. A finer, denser structure shears smaller particles at a higher rate, raising $D_{\max}$ (smaller particles scatter less light per unit mass of soot) and lowering τ (more contact points per stroke).

Results

Table 1 lists time constants for three stones. The grinding surface, not the ink stick, dominates.

The ordering is consistent across both parameters: the stone that saturates fastest also reaches the highest ceiling.

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Stone	τ (min)	$D_{\max}$
Duan (Laokeng)	4.1	0.94
Duan (Majiakeng)	6.8	0.87
She	9.2	0.81

Table 1: Saturation parameters by stone.

This coupling is what Equation 1 predicts if a single latent property — abrasive fineness — drives both effects. A Laokeng Duan stone reaches 90% of its ceiling in under ten minutes of grinding; the She stone requires more than twenty minutes to reach 90% of a lower ceiling. For a working calligrapher the practical gap is larger than the numbers suggest, because sessions are bounded by patience rather than by asymptotics.

It is worth stressing what the table does *not* show. It does not rank the stones aesthetically: She stones are prized for their eyebrow-streak figuring, and Chen and Nakamura (2021) note that provenance and market value track visual features far more closely than they track grinding performance. Nor does it capture durability, water retention, or the feel of the stroke, all of which enter a practitioner's choice.

Discussion

The folk claim survives quantification: the stone sets the ceiling. Two caveats bound the result. First, the measurements of Tanaka and Sato (2019) used standardized sticks; exotic formulations (lacquer-smoke soot, oil-soot blends) might yet interact with surface structure in ways the model excludes. Second, $D_{\max}$ is an optical ceiling, not a chemical one — prolonged grinding past saturation changes particle-size distribution without changing measured density, which may matter for brush behavior even where it does not matter for densitometry.

The identification argument in Equation 2 suggests a cheap field protocol for stone assessment: a one-minute grind and two density samples suffice to place a stone on the $(\tau, D_{\max})$ plane. We leave a validation of that protocol against the provenance methods of Chen and Nakamura (2021) to future work.

Limitations

Beyond the caveats above, the dataset itself is narrow: three stones, one per quarry tradition, each measured with a single stick formulation and a fixed water volume. Within-quarry variance is unreported, and Duan quarries in particular are known to yield stones of widely varying grade from the same seam. Temperature and humidity, both of which affect glue dissolution, were controlled in the laboratory but vary substantially in real studios. The model also treats grinding pressure and stroke rate as constant, whereas practitioners modulate both over a session; a richer model would let τ depend on applied work rather than on wall-clock time.

Conclusion

We restated a centuries-old craft claim as a two-parameter estimation problem and found that the data support it: the stone, not the stick, sets both the speed and the ceiling of ink saturation. The saturating-exponential form of Equation 1 fits all three stones tested, the fitted parameters order the stones consistently, and the coupling between τ and $D_{\max}$ points to a single latent driver in the surface microstructure. The practical upshot for a working calligrapher is simple, and it is the same advice the classical treatises give: invest in the stone.

Checklist

Opening this file and pressing preview should produce a two-column PDF with the AAAI running head, a numbered equation, a booktabs table, and a resolved citation to Tanaka and Sato (2019) in the references. The document should now run exactly two pages.

References

Chen, L.; and Nakamura, A. 2021. A Survey of Provenance Methods for She and Duan Inkstones. In *Proceedings of the 8th Conference on Heritage Materials Science*, 114–121. Heritage Press. Sample entry for rendering checks.

Mizuta, H. 1968. *The Scholar's Four Treasures: Stone, Stick, Brush, and Paper*. Kyoto: Kyoto University Press, 2nd edition.

Tanaka, Y.; and Sato, K. 2019. Ink Density Saturation Curves for Traditional Duan Inkstones. *Journal of East Asian Material Culture*, 12(3): 33–52.